Zero-Inflated Models

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Introduction

Zero-inflated models handle count data that contain more zeros than a Poisson or negative binomial distribution would predict — often called "structural zeros" because they arise from a process distinct from the count-generating process. The model is a mixture: one component decides whether each observation is a structural zero (always zero) or a sampling unit (could have been any non-negative count); the other component generates the count for the sampling units. Insurance claims, healthcare-utilisation counts, ecological abundance with absent species — all share this structure.

Prerequisites

A working understanding of Poisson and negative-binomial regression, logistic regression, and the conceptual distinction between mixture models and two-part hurdle models.

Theory

The zero-inflated mixture density is

$$\mathrm{P}(Y=0)=\pi+(1-\pi)f(0), \qquad \mathrm{P}(Y=y)=(1-\pi)f(y) \text{ for } y>0,$$

where π is the structural-zero probability (logistic regression on covariates) and f is the count distribution (Poisson or NB). The two sub-models have their own linear predictors and can use different covariate sets; observed zeros come from both the structural-zero process *and* the count-process zero.

The contrast with hurdle models is conceptual: zero-inflated treats some zeros as structurally always-zero, while hurdle treats *all* zeros as arising from one Bernoulli decision and the count distribution is truncated to be strictly positive.

Assumptions

Count outcome, two distinct processes generating zeros and non-zero counts, independent observations, and the chosen count distribution (Poisson or NB) appropriate for the non-structural component.

R Implementation

```r
library(pscl); library(glmmTMB)

set.seed(2026)
d <- data.frame(x = rnorm(300))
# Simulate with 40% structural zeros
pi_zero <- 0.4
d$y <- ifelse(rbinom(300, 1, pi_zero) == 1, 0,
              rpois(300, lambda = exp(0.5 + 0.6 * d$x)))

# Zero-inflated Poisson
```

```r
fit_zip <- zeroinfl(y ~ x | x, data = d)
summary(fit_zip)

# Zero-inflated negative binomial
fit_zinb <- zeroinfl(y ~ x | x, data = d, dist = "negbin")

AIC(fit_zip, fit_zinb)

# glmmTMB version
fit_tmb <- glmmTMB(y ~ x, ziformula = ~ x, data = d, family = poisson)
```

Output & Results

`summary(fit_zip)` returns two coefficient blocks: the count-component coefficients on the log-rate scale and the zero-inflation coefficients on the logit scale. `AIC()` compares ZIP against ZINB; substantially overdispersed counts favour ZINB strongly.

Interpretation

A reporting sentence: "A zero-inflated negative binomial model decomposed the observed zeros into structural and sampling sources: the count component estimated a rate ratio of 1.75 per unit increase in x (95 % CI 1.40 to 2.18, $p < 0.001$); the zero-inflation component estimated that a 1-unit increase in x reduced the odds of being a structural zero by 45 % (OR 0.55, 95 % CI 0.36 to 0.84, $p = 0.005$). ZINB was favoured over ZIP by AIC ($\Delta = 12$)." Always report both components and the framework rationale.

Practical Tips

- The Vuong test (`pscl::vuong`) compares zero-inflated to regular Poisson/NB but is often inconclusive; AIC and BIC are more reliable.
- Different covariates can drive the count and zero-inflation components; use `y ~ x_count | x_zero` to specify them separately.
- Hurdle models (`pscl::hurdle`) are an alternative when the scientific story is "did anything happen, and if so how much"; ZIP/ZINB fits "are some zeros structural while others are sampling".
- Overdispersed counts plus excess zeros typically warrant ZINB; pure overdispersion without structural zeros is just NB.
- Interpretation is cleaner in hurdle models when the two-stage scientific story is clear; ZIP/ZINB is preferable when zeros come from a genuine mixture.
- For random-effects extensions, `glmmTMB(..., ziformula = ~ ...)` adds clustered or hierarchical structure to either component.

R Packages Used

`pscl::zeroinfl()` and `hurdle()` for canonical implementations; `glmmTMB` for mixed-effects zero-inflated and hurdle models with cleaner overdispersion handling; `countreg` for additional count-model variants and rootogram diagnostics; `DHARMa` for residual diagnostics across zero-inflated families.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**

- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI